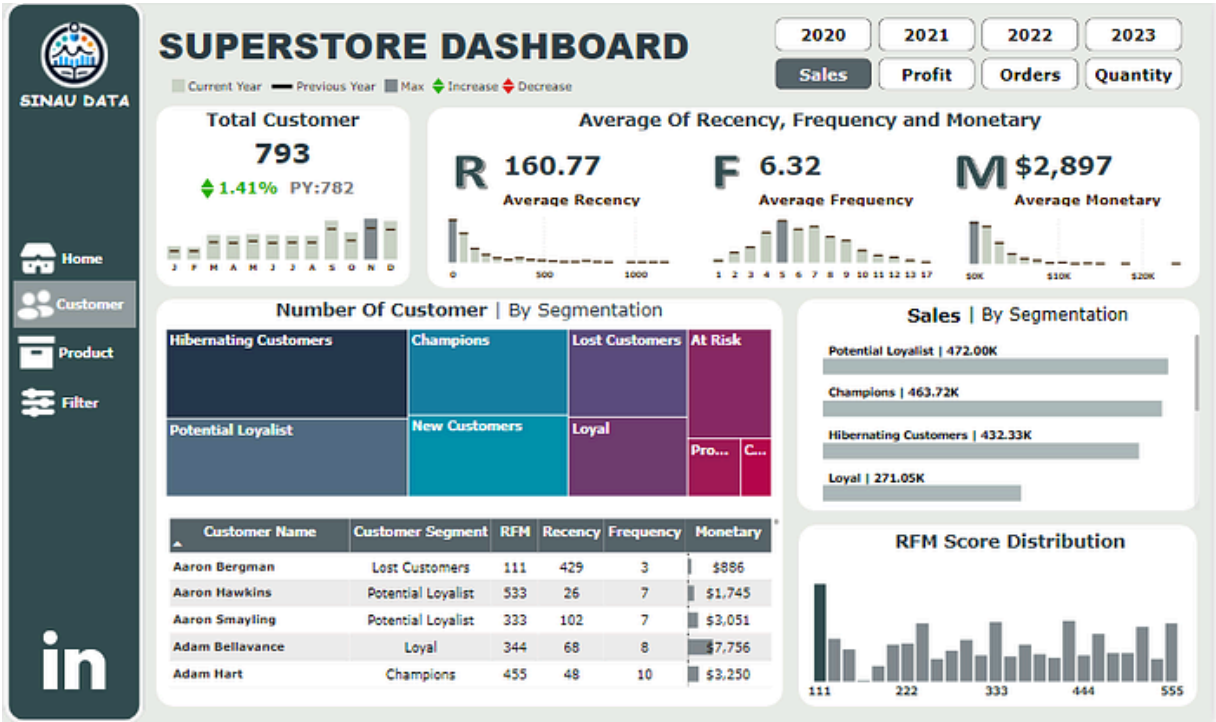


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Unleashing Customer Insights: A Guide to RFM Analysis Using Power BI

The RFM Customer Segmentation Analysis in Power BI equips businesses with the tools to transform raw data into actionable insights.



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Introduction

Understanding your customers is crucial for any business looking to thrive in today's competitive landscape. One powerful method for customer segmentation and analysis is the RFM model — Recency, Frequency, and Monetary. In this guide, we'll explore how to leverage the capabilities of Power BI to conduct RFM analysis using

source of information for extracting valuable customer insights.

What is RFM Analysis?

RFM analysis is a data-driven marketing technique that segments customers based on their transaction history. It evaluates three key metrics:

- **Recency (R):** How recently a customer made a purchase.
- **Frequency (F):** How often a customer makes a purchase.
- **Monetary (M):** The total amount spent by a customer.

By categorizing customers into different segments based on these metrics, businesses can tailor their marketing strategies to better meet the specific needs of each segment. You can download the RFM segment score table to easily interpret and analyze your customer segments. [Click here to download the RFM segment score table.](#)

Segment	Scores
Champions	555, 554, 544, 545, 454, 455, 445
Loyal	543, 444, 435, 355, 354, 345, 344, 335
Potential Loyalist	553, 551, 552, 541, 542, 533, 532, 531, 452, 451, 442, 441, 431, 453, 433, 432, 423, 353, 352, 351, 342, 341, 333, 323
New Customers	512, 511, 422, 421, 412, 411, 311
Promising	525, 524, 523, 522, 521, 515, 514, 513, 425, 424, 413, 414, 415, 315, 314, 313
Need Attention	535, 534, 443, 434, 343, 334, 325, 324
About To Sleep	331, 321, 312, 221, 213, 231, 241, 251
At Risk	255, 254, 245, 244, 253, 252, 243, 242, 235, 234, 225, 224, 153, 152, 145, 143, 142, 135, 134, 133, 125, 124
Cannot Lose Them	155, 144, 214, 215, 115, 114, 113
Hibernating Customers	332, 322, 231, 241, 251, 233, 232, 223, 222, 132, 123, 122, 212, 211
Lost Customers	111, 112, 121, 131, 141, 151

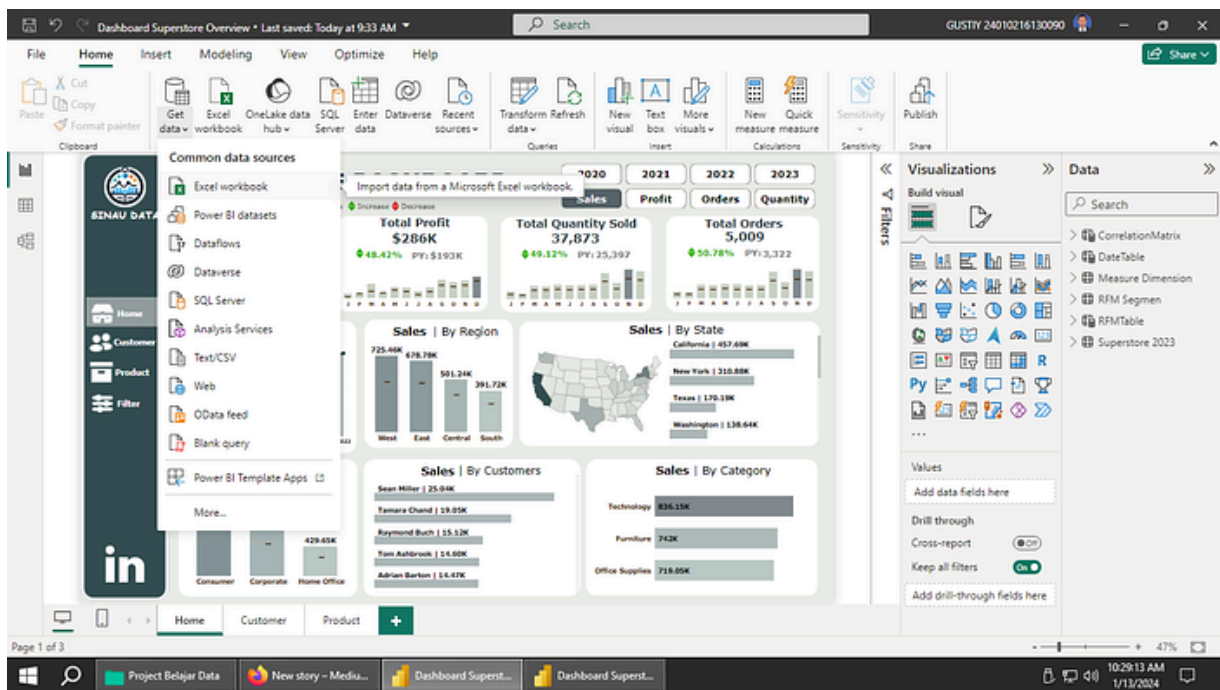
Table Segment Customers and Score RFM

Loyal	Spend good money with us often. Responsive to promotions
Potential Loyalist	Recent customers, but spent a good amount and bought more than once.
New Customers	Bought most recently, but not often.
Promising	Recent shoppers, but haven't spent much.
Need Attention	Above average recency frequency and monetary values. May not have bought very recently though.
About To Sleep	Below average recency frequency and monetary values Will lose them if not reactivated
At Risk	Spent big money and purchased often. But long time ago. Need to bring them back!
Cannot Lose Them	Made biggest purchases, and often. But haven't returned for a long time.
Hibernating Customers	Last purchase was long back, low spenders and low number of orders.
Lost Customers	Lowest recency, frequency and monetary scores

Detailed Information About The Segment

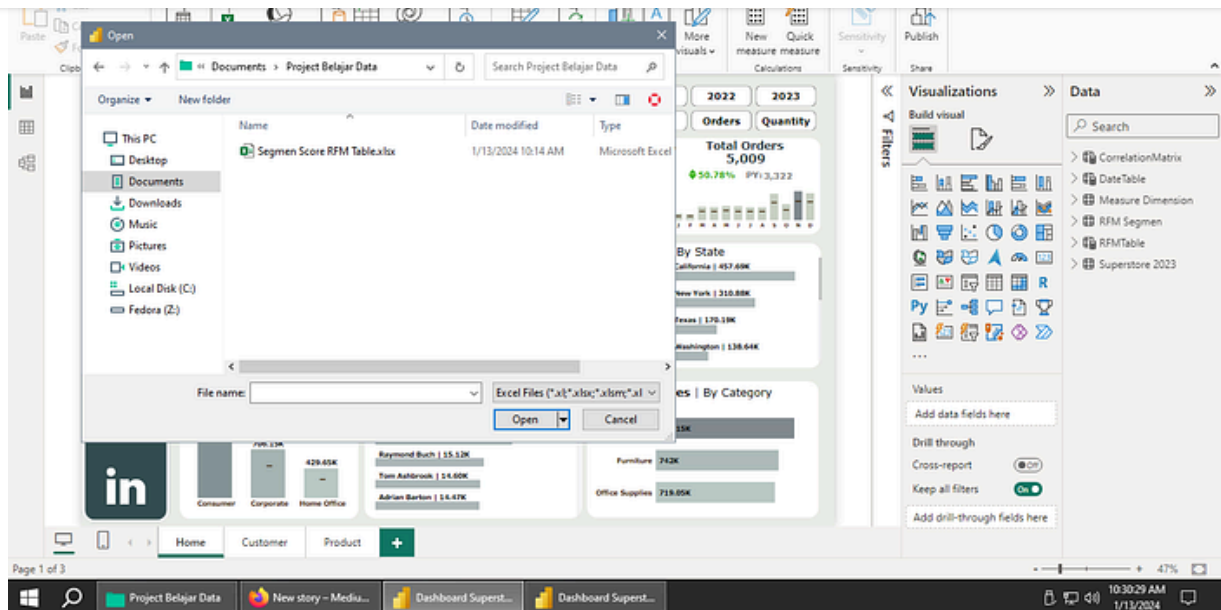
Data Loading

Download the Superstore 2023 dataset from data.world and load it into Power BI. 1. **Get Data:** Click "Get Data" on the "Home" tab in Power BI Desktop.



Get Data

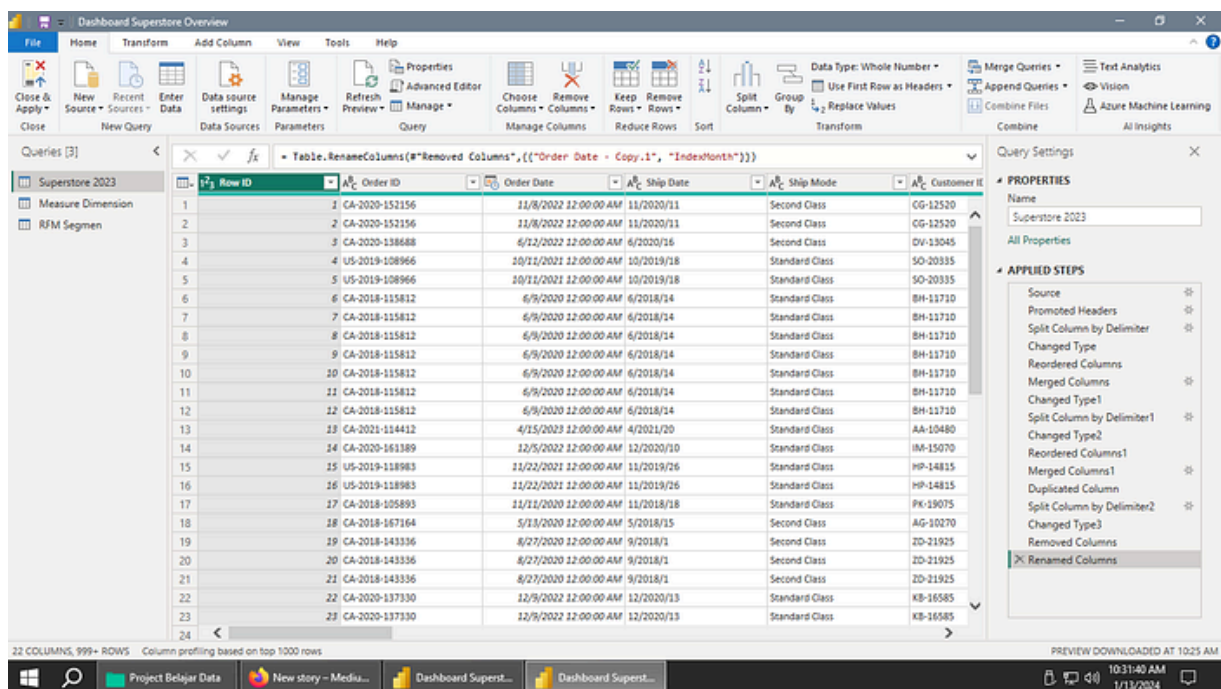
2. **Choose Data Source:** Select your data source (Excel, Text/CSV, Database, Web, etc.).



Choose Data Source

3. Connect to Your Data: - Provide connection details or choose your file. - Click "OK" or "Connect" to proceed.

4. Load Data into Power BI: - Power BI will preview your data. - Transform and clean if needed, then click "Close & Apply".



Transform Data

Create Date Measures

Transform the 'Order Date' into a Date/Time format and create measures for Last Transaction Date, R Value (Recency), F Value (Frequency), and M Value (Monetary).

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```
LastTransactionDate = MAXX(FILTER('Superstore 2023', 'Superstore 2023'[Order Date] <= TODAY()), 'Superstore 2023'[Order Date])

R value = DATEDIFF('Superstore 2023'[LastTransactionDate], TODAY(), DAY)

F value = CALCULATE(DISTINCTCOUNT('Superstore 2023'[Order ID]), ALL('Superstore 2023'))

M value = SUM('Superstore 2023'[Sales])
```

Create RFM Table Using DAX

1. Open the "Model" view in Power BI Desktop.
2. Click "New Table" on the "Model" tab.
3. Enter the following DAX code:

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```
RFMTable =
SUMMARIZE(
    'Superstore 2023',
    'Superstore 2023'[Customer ID],
    'Superstore 2023'[Customer Name],
    "Recency", [R value],
    "Frequency", [F value],
    "Monetary", [M value]
)
```

This DAX code uses the SUMMARIZE function to create a new table named RFMTable. It summarizes the Superstore 2023 data, grouping

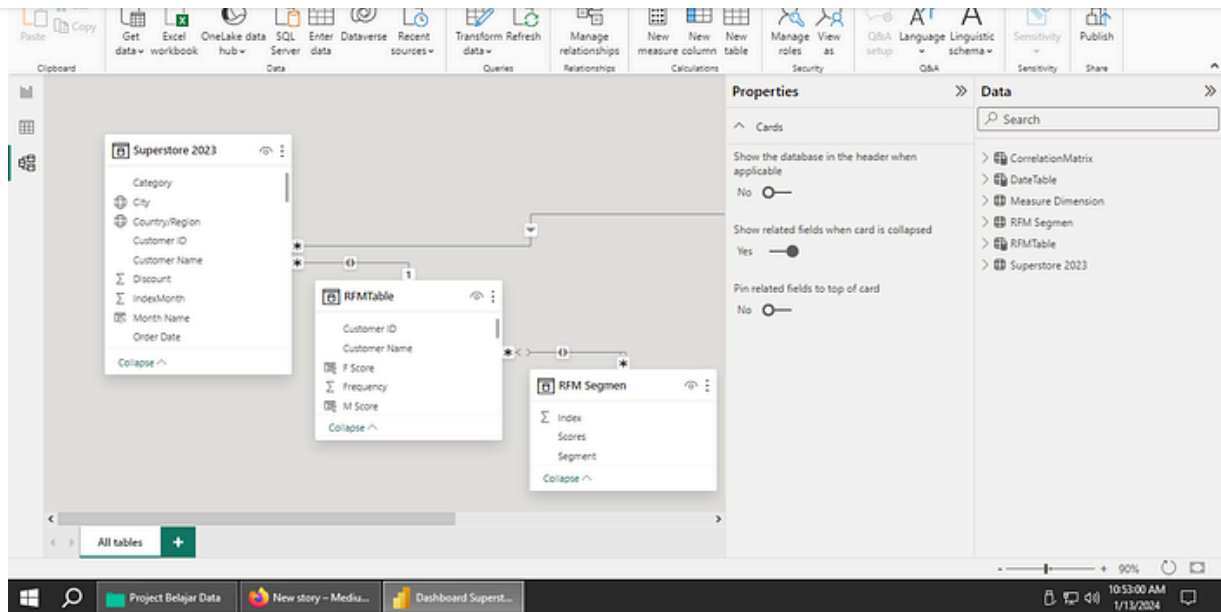
Customer ID	Recency	Frequency	Monetary	R Score	F Score	M Score	RFM	Customer Name	Segment	Recency (bins)	Monetary (bins)
BN-11515	23	5	\$1,459,344	5	2	2	522	Bradley Nguyen	Promising	0	
PL-18925	29	5	\$239,481	5	2	2	522	Paul Lucas	Promising	0	
RW-19690	93	5	\$3,495,646	3	2	2	322	Robert Waldorf	Hibernating Customers	61	
SM-20320	93	5	\$25,043.05	3	2	2	322	Sean Miller	Hibernating Customers	61	
BT-11530	105	5	\$2,684,492	3	2	2	322	Bradley Talbot	Hibernating Customers	61	
OT-18730	79	5	\$3,818,624	3	2	2	322	Olivera Tooh	Hibernating Customers	61	
JC-16105	71	5	\$858,576	3	2	2	322	Julie Creighton	Hibernating Customers	61	
VF-21715	27	5	\$847,944	5	2	2	522	Vicky Freymann	Promising	0	
CC-12370	57	5	\$12,129,072	4	2	2	422	Christopher Conant	New Customers	0	
TS-21430	186	5	\$2,054.14	2	2	2	222	Tom Stivers	Hibernating Customers	183	
JM-15580	14	5	\$303,948	5	2	2	522	Jill Matthias	Promising	0	
BP-11230	53	5	\$1,181.49	4	2	2	422	Benjamin Patterson	New Customers	0	
DB-12970	255	5	\$2,839,2272	1	2	2	122	Darren Budd	Hibernating Customers	244	
DV-13045	33	5	\$1,119,483	5	2	2	522	Darrin Van Huff	Promising	0	
CD-12790	413	5	\$1,860,732	1	2	2	122	Cynthia Delaney	Hibernating Customers	366	
CA-12055	415	5	\$1,679,724	1	2	2	122	Cathy Armstrong	Hibernating Customers	366	

RFM Table

Establishing Relationships

To enrich our analysis, we need to establish a relationship between the Superstore 2023 table, the RFM Table and The RFM Segment Table. This is crucial for seamlessly integrating RFM insights into our main dataset.

1. In Power BI Desktop, navigate to the "Model" view.
2. Identify common fields between the Superstore 2023 table and the RFM Table, such as 'Customer ID'.
3. Drag and drop the 'Customer ID' field from both tables onto the canvas to create a relationship.



Establishing Relationships

RFM Measures and Customer Segmentation

Categorize Recency, Frequency, and Monetary into segments and create a dynamic segmentation for each customer. Create calculated column using this DAX.

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```
R Score = SWITCH (
    TRUE (),
    [Recency] <= PERCENTILE.INC ( 'RFMtable'[Recency], 0.20 ), "5",
    [Recency] <= PERCENTILE.INC ( 'RFMtable'[Recency], 0.40 ), "4",
    [Recency] <= PERCENTILE.INC ( 'RFMtable'[Recency], 0.60 ), "3",
    [Recency] <= PERCENTILE.INC ( 'RFMtable'[Recency], 0.80 ), "2",
    "1"
)

F Score = SWITCH (
    TRUE (),
    [Frequency] <= PERCENTILE.INC ( 'RFMTable'[Frequency], 0.20 ),
    [Frequency] <= PERCENTILE.INC ( 'RFMTable'[Frequency], 0.40 ),
    [Frequency] <= PERCENTILE.INC ( 'RFMTable'[Frequency], 0.60 ),
    [Frequency] <= PERCENTILE.INC ( 'RFMTable'[Frequency], 0.80 ),
    "5"
)

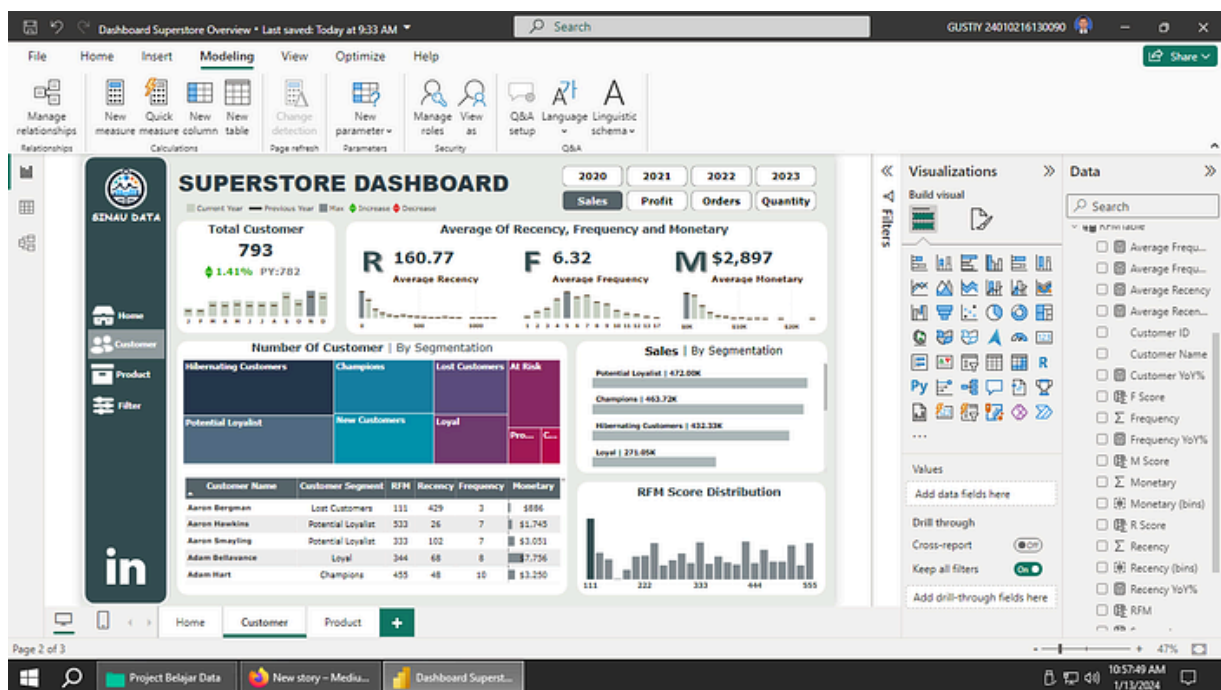
M Score = SWITCH (
```

```
[Frequency] <= PERCENTILE.INC ( 'RFMTable' [Frequency], 0.60 ),
[Frequency] <= PERCENTILE.INC ( 'RFMTable' [Frequency], 0.80 ),
"5"
)
```

```
RFM = 'RFMTable' [R Score]& 'RFMTable' [F Score]&'RFMTable' [M Score]
Segment = LOOKUPVALUE('RFM Segmen' [Segment], 'RFM Segmen' [Scores], R
```

Visualization and Analysis

Transitioning to the Report view, leverage the Power BI canvas to craft engaging visuals. Utilize charts, tables, and slicers to dynamically analyze and interpret RFM segments. This step empowers you to uncover patterns, identify high-value customers, and tailor marketing strategies accordingly.



RFM Analysis Dashboard

Insight From The Dashboard:

1. **High Potential Customers:** - **Segments:** Champions, Potential Loyalists, and Loyal customers. - **Insight:** Focus on retaining and

2. **Engagement Opportunities:** - **Segments:** New Customers and Promising. - **Insight:** Implement strategies to increase engagement and order frequency to maximize their long-term value.
3. **At-Risk Customers:** - **Segment:** At Risk. - **Insight:** Proactively address potential disengagement to prevent customer churn and sustain profitability.
4. **Reactivation Strategies:** - **Segment:** Hibernating Customers and Lost Customers. - **Insight:** Develop targeted reactivation strategies to re-engage with these customers and win back their business.
5. **Small but Valuable:** - **Segment:** Cannot Lose Them. - **Insight:** Despite a small customer base, this segment contributes significantly. Strengthen loyalty through personalized strategies.

In analyzing the RFM Customer Segmentation, a notable revelation emerges with the largest customer segment being the Hibernating Customers, encompassing a substantial 170 individuals. Despite their numerical dominance, this segment displays lower engagement levels, reflected in their modest total sales of \$432,331.22 and a profit margin of \$57,036.52. Alongside this, there's a clear focus on retaining and engaging with high-value segments like Champions, Potential Loyalists, and Loyal customers.

Conclusion

In conclusion, the RFM Customer Segmentation project in Power BI equips businesses with the tools to transform raw data into actionable insights. By understanding customer behavior through Recency, Frequency, and Monetary metrics, organizations can